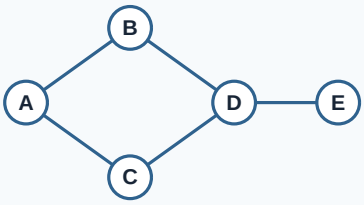
BFS and DFS: three worked examples

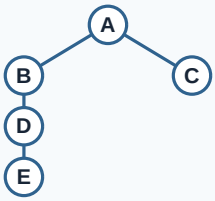
Start at A and consider neighbors alphabetically. BFS marks on enqueue; DFS enters the first unvisited neighbor.

1. Diamond and tail

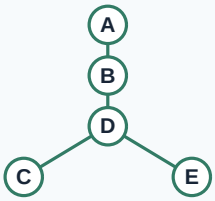
Input graph



BFS tree



DFS tree

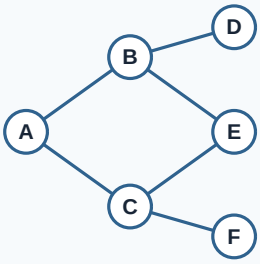


Frontier by level:
[A] [B,C] [D] [E] []

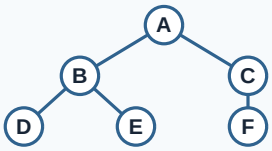
Visit order:
A, B, D, C, E

2. Shared neighbor

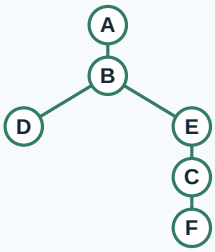
Input graph



BFS tree



DFS tree

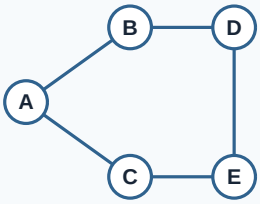


Frontier by level:
[A] [B,C] [D,E,F] []

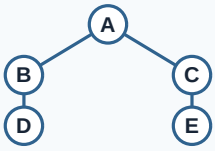
Visit order:
A, B, D, E, C, F

3. Two routes

Input graph



BFS tree



DFS tree

Frontier by level:
[A] [B,C] [D,E] []

Visit order:
A, B, D, E, C